CASCADE App Summary

Repository: /Users/patrickgallowaypro/Documents/PROJECTS/CASCADE

What It Is

CASCADE is a TypeScript monorepo for building, running, and observing AI workflow playbooks across workspaces. It includes a Next.js operator console, a Node worker, a shared runtime engine, and Prisma-backed data models.

Who It's For

Primary persona: workflow operators and engineering teams managing multi-step AI automations for client/workspace operations.

Explicit persona statement: Not found in repo.

What It Does

- Manages workspaces, playbooks, versions, triggers, runs, secrets, templates, and agent pages in the console UI.
- Executes workflow runs with step status tracking, token usage, and cost accounting per run/step.
- Supports node types for LLM, HTTP, Slack, branch, transform, and wait execution paths.
- Validates playbook definitions with shared Zod schemas and runtime parsing utilities.
- Enforces budget guardrails (token and cost caps) in runtime utilities and workspace settings.
- Provides passwordless magic-link auth via Supabase Auth and email delivery via Resend.

How It Works (Repo-Evidenced Architecture)

• **Console (apps/console)** -> Next.js dashboard + API routes create/manage runs, playbooks, triggers, and secrets.

• **Runtime (packages/runtime)** -> parse/validate playbook JSON, build node map, render templates, evaluate expressions/branches, check guardrails, compute costs.

• **Worker (apps/worker)** -> pg-boss consumer for *execute-run*; loads run data from Prisma, executes nodes, writes run-step outcomes.

• **Database (packages/db/prisma)** -> PostgreSQL models for User, Workspace, Playbook/Version, Trigger, Run/RunStep, Guardrail, UsageLog, Secret.

• **External services** -> OpenAI (LLM outputs), Slack API, Supabase (auth/data), Resend (magic-link email).

• **Current run path in console API** -> asynchronous in-process step execution is present; direct queue handoff from console route is Not found in repo.

How To Run (Minimal)

- Install deps: `pnpm install`
- Configure env: `cp .env.example .env` then fill required values (DB, Supabase, OpenAI, secrets).
- Initialize DB: `pnpm db:generate && pnpm db:push` (optional seed: `pnpm db:seed`).
- Start dev: `pnpm dev` (console runs on `http://localhost:3001`; worker runs in watch mode).